



Real-time oceanic PWV sensing using BeiDou-3 PPP-B2b and low-cost GNSS devices

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Abstract

Real-time sensing of precipitable water vapor (PWV) using GNSS is critical for atmospheric monitoring. Conventional real-time precise point positioning (PPP) reliant on the IGS real-time service (RTS) remains limited in oceanic regions due to terrestrial internet dependencies. BeiDou's satellite-based PPP-B2b service overcomes this limitation, enabling real-time PWV sensing over the ocean. However, its relatively lower accuracy has hindered broader application. Additionally, the potential of combining PPP-B2b with low-cost GNSS devices for oceanic PWV sensing remains insufficiently explored. This study presents a practical and cost-effective approach for real-time oceanic PWV sensing by jointly using PPP-B2b and low-cost GNSS devices. To address the inherent errors in PPP-B2b corrections, we develop a refined PPP strategy based on ambiguity reparameterization and optimized stochastic modeling. Unlike conventional PPP that treats ambiguities as constants, this method models them as time-varying random-walk processes within an optimized range of process-noise values. This allows for adaptive absorption of PPP-B2b correction errors, thereby improving the accuracy of tropospheric estimates. Validation through land-based static and kinematic experiments demonstrates accuracy improvements of 34 and 24%, respectively, in tropospheric estimates. When applied to IGS-RTS corrections, the refined method achieves a 10% accuracy improvement in real-time tropospheric estimation. Two oceanic campaigns further validate its operational feasibility, yielding a PWV accuracy improvement of over 30% during Cruise 1 and 47% during Cruise 2 compared to conventional PPP. This framework offers a seamless, accurate and cost-effective solution for real-time tropospheric and PWV sensing, benefiting weather forecasting and climate monitoring.

Keywords BeiDou navigation satellite system · Real · Time precise point positioning (PPP) · PPP · B2b · Oceanic precipitable water vapor · Low · Cost GNSS devices

1 Introduction

The ocean is the largest source of atmospheric water vapor, contributing approximately 86% of global

evaporation (Schmitt et al. 1995). Real-time sensing of oceanic precipitable water vapor (PWV) is critical for precise and timely global atmospheric analyses, particularly for short-term severe weather forecasting (Huang et al. 2021; Gong et al. 2021; Schneider et al. 2010; Zheng et al. 2022). Several methods for oceanic PWV sensing have been developed, utilizing both spaceborne (Du et al. 2014; Gao et al. 2003; Xu and Liu 2024) and shipborne sensors (Fujita et al. 2008; Wang et al. 2019), but each with its own limitations. For instance, spaceborne microwave sensors can offer broad coverage of PWV measurements but are heavily affected by weather conditions (He and Liu 2019; Jiang et al. 2024). In contrast, GNSS technology offers continuous, all-weather PWV observations and is increasingly recognized as a promising method for shipborne PWV sensing (Männe et al. 2021; Shoji et al. 2016; Wang et al. 2019).

Using shipborne GNSS for real-time oceanic PWV sensing presents additional challenges compared to

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land-based GNSS stations. These challenges stem from dynamic positioning conditions, limited telecommunications infrastructure, and complex observation environment in oceanic regions (Rocken et al. 2005; Shoji et al. 2016; Wei et al. 2023). Nevertheless, previous studies have demonstrated the feasibility of using shipborne GNSS for oceanic PWV sensing.

Early efforts, such as those by Chadwell et al. (2001), employed the GPS relative positioning techniques. However, this approach's dependence on reference stations limited its application in remote oceanic regions. In contrast, GNSS precise point positioning (PPP) strategy, which operates independently of reference stations, has become the dominant method for shipborne PWV sensing. For instance, Rocken et al. (2005) demonstrated the potential of postprocessed PPP for oceanic PWV sensing, achieving an accuracy of 2.8 mm. Shoji et al. (2016) further explored the PPP strategy, employing file-based GNSS satellite ephemeris to retrieve PWV with an accuracy of 3.4 ~ 5.4 mm when compared to radiosonde observations. Wang et al. (2019) extended the application of postprocessed PPP to multi-GNSS observations for shipborne PWV sensing, highlighting the advantages of multi-GNSS solutions over GPS-only approaches. Männel et al. (2021) leveraged postprocessed PPP to retrieve polar PWV using shipborne GNSS data. More recently, Wu et al. (2022) explored the use of simulated real-time PPP for oceanic PWV sensing, achieving an accuracy of 4.0 mm. These findings collectively underscore the evolving capabilities of shipborne GNSS for oceanic PWV sensing and establish a foundation for further advancements in real-time applications.

Real-time PPP requires precise GNSS satellite orbit and clock corrections, which are typically provided by the International GNSS Service (IGS) real-time service (RTS) (Hadas et al. 2020; Lee et al. 2013; Li et al. 2014; Lu et al. 2015; Yuan et al. 2014). However, the reliance on terrestrial internet infrastructure limits the applicability of IGS-RTS in remote oceanic regions. Some studies have examined the feasibility of real-time PPP using broadcast ephemeris (Carlin et al. 2021; Chen et al. 2022; Pan et al. 2024), but the accuracy achieved is inferior to that of PPP solutions utilizing precise corrections. Satellite-based PPP services, which broadcast precise corrections via satellite, offer a promising solution for real-time high-accuracy PWV sensing in areas without internet connectivity.

The BeiDou Navigation Satellite System (BDS-3) provides a novel satellite-based PPP service, known as PPP-B2b, which broadcasts real-time precise orbit and clock corrections for GPS and BDS-3 satellites via the B2b signals of Geostationary Earth Orbit (GEO) satellites (Tang et al. 2022; Yang et al. 2019). Recently, Tao et al. (2025) present a new approach by extracting phase observable-specific bias compatible with the PPP-B2b service, demonstrating potential

for improving its efficiency and performance. Additionally, the Galileo High Accuracy Service (HAS) offers another satellite-based PPP service, providing precise corrections for GPS and Galileo satellites (Naciri et al. 2023). Researches have explored the integration of PPP-B2b and Galileo HAS to enhance PPP performance (Pan et al. 2025; Wei et al. 2025).

Nevertheless, real-time orbit and clock corrections may introduce systematic biases that degrade PPP performance (Hadas et al. 2020; Ji et al. 2023). For PPP-B2b, such biases are typically more pronounced because its corrections are generated from a limited tracking network, resulting in lower accuracy than that of IGS-RTS (Tang et al. 2022; Xu et al. 2023). While several studies have evaluated the potential of PPP-B2b for estimating zenith tropospheric delay (ZTD) and PWV, Yang et al. 2023; Zhou et al. 2024 most of these investigations have been conducted using postprocessed or simulated real-time configurations, typically relying on land-based high-grade GNSS stations. Few studies have validated PPP-B2b under true real-time conditions. Moreover, the relatively lower accuracy of PPP-B2b corrections remains a bottleneck for achieving high-quality real-time PWV retrieval, and this challenge has not yet been adequately addressed. Meanwhile, low-cost GNSS devices, ideal for deployment on large-scale mobile marine platforms such as buoys, are gaining popularity for dense PWV sensing (Marut et al. 2022). However, the integration of PPP-B2b with such low-cost devices for real-time oceanic PWV monitoring remains insufficiently explored.

This study aims to advance real-time oceanic PWV sensing by jointly using PPP-B2b and low-cost GNSS devices. To address the limitations of PPP-B2b correction accuracy, we develop a refined PPP method that effectively mitigates correction errors, thereby improving the accuracy of real-time ZTD and PWV sensing in oceanic environments.

2 Methodology

We employed the real-time PPP technique to process real-time GNSS data streams for estimating real-time ZTD. These ZTD estimates were subsequently converted to PWV using auxiliary atmospheric parameters, namely, the zenith hydrostatic delay (ZHD) and weighted mean temperature. Specifically, ZHD was computed using the Saastamoinen model, while weighted mean temperature was derived from the Bevis empirical model (Bevis et al. 1992), with in-situ pressure and temperature measurements as inputs.

2.1 Conventional PPP method

We adopt the ionosphere-free (IF) combination for PPP. By leveraging precise satellite orbit and clock products, along

with an a priori tropospheric delay correction, the IF pseudorange and carrier-phase observation equations can be written as:

$$\begin{aligned}\Delta P_{\text{if}}^s &= -\mathbf{e}_r^s \cdot d\mathbf{x} + c \cdot dt_r + M_r^s \cdot dZ + v_p, \\ \Delta \Phi_{\text{if}}^s &= -\mathbf{e}_r^s \cdot d\mathbf{x} + c \cdot dt_r + M_r^s \cdot dZ + \lambda_{\text{if}} \cdot N_{\text{if}}^s + v_\Phi,\end{aligned}\quad (1)$$

where ΔP_{if}^s and $\Delta \Phi_{\text{if}}^s$ are the observed minus computed IF pseudorange and carrier-phase for satellite s ; $\mathbf{e}_r^s$ is the line-of-sight unit vector; $d\mathbf{x}$ is the position increment vector; C is the speed of light; dt_r is the receiver clock error; dZ is the correction to the a priori zenith tropospheric delay; M_r^s is the tropospheric mapping function. N_{if}^s is the IF ambiguity; λ_{if} is the IF-combination wavelength; and v_p and v_Φ are the measurement noise terms.

The unknown variables, $d\mathbf{x}$, dt_r , dZ , and N_{if}^s , formed a state vector $\mathbf{X} = [d\mathbf{x}, dt_r, dZ, N_{\text{if}}^s]^T$, which is estimated via an extended Kalman filter (EKF). In the conventional PPP, the ambiguity N_{if}^s is treated as time-invariant with zero process noise in the EKF prediction step:

$$\mathbf{X}_{k|k-1} = \mathbf{\Gamma}_{k|k-1} \cdot \mathbf{X}_{k-1} + \begin{bmatrix} \delta_{d\mathbf{x}} \\ \delta_{dt_r} \\ \delta_{dZ} \\ \delta_{N_{\text{if}}^s} \end{bmatrix} \cdot \Delta t, \text{ with } \delta_{N_{\text{if}}^s} = 0 \quad (2)$$

where $\mathbf{\Gamma}_{k|k-1}$ is the state transition matrix, k denotes the epoch number, $\mathbf{X}_{k|k-1}$ is the predicted state at epoch k ; and $\delta_{d\mathbf{x}}$, δ_{dt_r} , δ_{dZ} , and $\delta_{N_{\text{if}}^s}$ represent the process-noise terms for the receiver coordinates, receiver clock error, zenith tropospheric delay, and ambiguity, respectively, with Δt being the time interval between epochs. The state is subsequently updated using the innovation vector within the EKF framework.

This conventional PPP formulation implicitly assumes error-free satellite orbit and clock products, an assumption that does not hold when using real-time corrections.

2.2 Refined real-time PPP method

Real-time orbit and clock corrections (e.g., from PPP-B2b) inevitably contain residual errors. These residuals, denoted E_r^s , typically range from a few centimeters to several decimeters (Tang et al. 2022) and exhibit temporal smoothness and satellite-specific variations (Xu et al. 2023). The IF observation equations with imperfect real-time corrections become:

$$\begin{aligned}\Delta P_{\text{if}}^s &= -\mathbf{e}_r^s \cdot d\mathbf{x} + c \cdot dt_r + M_r^s \cdot dZ + E_r^s + v_p, \\ \Delta \Phi_{\text{if}}^s &= -\mathbf{e}_r^s \cdot d\mathbf{x} + c \cdot dt_r + M_r^s \cdot dZ + \lambda_{\text{if}} \cdot N_{\text{if}}^s + E_r^s + v_\Phi,\end{aligned}\quad (3)$$

One option is to augment the state vector with explicit per-satellite E_r^s parameters and estimate them directly (Gunning et al. 2019; Xu et al. 2023). While effective, this approach increases the state dimensionality by the number of tracked satellites and complicates the filter.

An alternative, adopted here and motivated by the correlation analysis below, is to exploit the structural coupling between E_r^s and N_{if}^s (as illustrated in Sect. “Correlation analysis and theoretical justification for the refined PPP method”). The E_r^s for satellite s can be decomposed as:

$$E_r^s = \lambda_{\text{if}} \cdot \Delta N_{\text{if}}^s + \xi_{\text{res}} \quad (4)$$

where ΔN_{if}^s is the ambiguity-equivalent component that can be absorbed into N_{if}^s , and ξ_{res} is small residual. Substituting (4) into (3) yields augmented residual terms and illustrates that the effect of E_r^s can be largely represented by permitting the ambiguity to evolve as a slow random walk. Hence, rather than estimating E_r^s explicitly, we allow N_{if}^s to have non-zero process noise:

$$N_{\text{if}}^s(k) = N_{\text{if}}^s(k-1) + \omega_{N_{\text{if}}^s}(k), \quad \omega_{N_{\text{if}}^s} \sim \mathcal{N}(0, Q_{N_{\text{if}}^s}) \quad (5)$$

with $Q_{N_{\text{if}}^s}$ tuned to reflect the typical E_r^s temporal variability. This refined stochastic modeling lets the EKF adaptively absorb slowly varying E_r^s without increasing the state dimension. As demonstrated through a comparative analysis in the Appendix A, this simplified approach achieves comparable mitigation performance to the explicit parameter estimation method.

2.3 Correlation analysis and theoretical justification for the refined PPP method

The refined PPP strategy in Sect. “Refined real-time PPP method” implicitly relies on a key structural property of the PPP model, namely, the strong satellite-specific coupling between N_{if}^s and E_r^s . This coupling is the theoretical basis that enables N_{if}^s to effectively absorb E_r^s , while preventing their leakage into the tropospheric estimates and other global parameters. To substantiate this prerequisite, we conduct the following two-part analysis:

First, we derive the expected correlation structure from the linearized PPP design matrix, demonstrating the inherently dominant coupling between N_{if}^s and E_r^s . Second, to validate the theory under realistic conditions, we conduct an empirical correlation analysis using real GPS and BDS-3 data, in which state-wise correlations are quantified based on the covariance matrix of the PPP solution. Together, these analyses substantiate the theoretical basis of the refined PPP method.

2.3.1 (a) Theoretical correlation characteristics among E_r^s and PPP parameters.

From the linearized phase observation equation with E_r^s (Eq. (3)):

$$\Delta\Phi_{if}^s = -e_r^s \cdot dx + c \cdot dt_r + M_r^s \cdot dZ + \lambda_{if}^s \cdot N_{if}^s + E_r^s + v_\Phi, \quad (6)$$

The sensitivities of the observation to N_{if}^s and E_r^s are:

$$\frac{\partial \Delta\Phi_{if}}{\partial N_{if}^s} = \lambda_{if}^s, \quad \frac{\partial \Delta\Phi_{if}}{\partial E_r^s} = 1 \quad (7)$$

i.e., both are additive, satellite-specific terms that enter the equation with constant coefficients. Consequently, the corresponding columns of the PPP design matrix are nearly collinear, resulting in a near-perfect correlation between N_{if}^s and E_r^s . Therefore, the correlation coefficient ρ between these two parameters is approximately:

$$|\rho(N_{if}^s, E_r^s)| \approx 1 \quad (8)$$

In contrast, other PPP parameters, including receiver coordinates ($d\mathbf{x}_r^s$), receiver clocks (dt_r), and tropospheric delays (dZ), affect the observations through geometry or mapping factors:

$$\frac{\partial \Delta\Phi_{if}}{\partial d\mathbf{x}} = -\mathbf{e}_r^s, \quad \frac{\partial \Delta\Phi_{if}}{\partial dt_r} = c, \quad \frac{\partial \Delta\Phi_{if}}{\partial dZ} = M_r^s \quad (9)$$

These columns vary across satellites through $\mathbf{e}_r^s$ or M_r^s , representing global or geometry-dependent signatures rather than satellite-specific offsets. Consequently, the structural collinearity that exists between E_r^s and N_{if}^s does not exist between E_r^s and $d\mathbf{x}$ or dZ ; their correlations are therefore inherently weaker, yielding:

$$\begin{aligned} |\rho(d\mathbf{x}, E_r^s)| &\ll |\rho(N_{if}^s, E_r^s)|, \quad |\rho(dt_r, E_r^s)| \\ &\ll |\rho(N_{if}^s, E_r^s)|, \quad |\rho(dZ, E_r^s)| \ll |\rho(N_{if}^s, E_r^s)| \end{aligned} \quad (10)$$

This structural reasoning justifies the choice of absorbing E_r^s predominantly into the ambiguity states via modest non-zero process noise.

2.3.2 (b) Correlation analysis using real PPP covariance matrix

To empirically validate the above theoretical analysis, we further investigated the covariance structure of a PPP solu-

tion derived from real GNSS observations. A full-day dataset collected at the DHSP station on 17 November 2025 was processed using a modified PPP model in which the E_r^s parameters were explicitly estimated. The state covariance matrix $\mathbf{P}$ at the final epoch was then extracted for analysis. The interpreted state vector is given by:

$$\mathbf{X} = \begin{bmatrix} dx, dy, dz, dt_{rg}, dt_{rc}, dZ, \underbrace{N_{G1} \cdots N_{Gm}, N_{C1} \cdots N_{Cm}}_{\text{GPS/BDS - 3 ambiguities}}, \\ \underbrace{E_{G1} \cdots E_{Gm}, E_{C1} \cdots E_{Cm}}_{\text{GPS/BDS - 3 } E_r^s} \end{bmatrix}^T \quad (11)$$

The corresponding correlation matrix was computed as:

$$\rho_{ij} = \frac{P_{ij}}{\sqrt{P_{ii}P_{jj}}} \quad (12)$$

Figure 1 presents the resulting heatmap. A pronounced satellite-wise diagonal pattern is evident, indicating strong correlations between each ambiguity parameter N_{if}^s and its corresponding E_r^s . This is consistent with the theoretical collinearity in Eq. (7) and (8). In contrast, coordinate and tropospheric delay parameters exhibit relatively weak correlations with E_r^s , reflecting their global and geometry-dependent nature, as described in Eq. (9) and (10). These empirical observations align with the theoretical analysis in Sect. “Correlation analysis and theoretical justification for the refined PPP method” (a) and confirming that ambiguity parameters are primarily responsible for absorbing E_r^s .

2.4 Real-time PPP processing strategies

The success of the refined PPP method depends on selecting an appropriate process-noise $\delta_{N_{if}^s}$ for the ambiguity. To investigate this, we conducted an empirical sensitivity analysis by varying $\delta_{N_{if}^s}$ from 0 to 21.0 mm/ $\sqrt{h}$ in increments of 3.0 mm/ $\sqrt{h}$. Eight schemes (S0-S7) were tested, with S0 representing the conventional PPP method ($\delta_{N_{if}^s} = 0.0$ mm/ $\sqrt{h}$). The scheme designations are summarized in Table 1. This design enables a comparative evaluation of filter performance under different stochastic assumptions.

The detailed configurations adopted for real-time PPP processing in both static and kinematic modes are summarized in Table 2. Notably, the tropospheric gradient model of Chen et al. (1997) is applied only in static mode, as azimuthal asymmetries in the troposphere cannot be reliably estimated when the antenna is in motion.

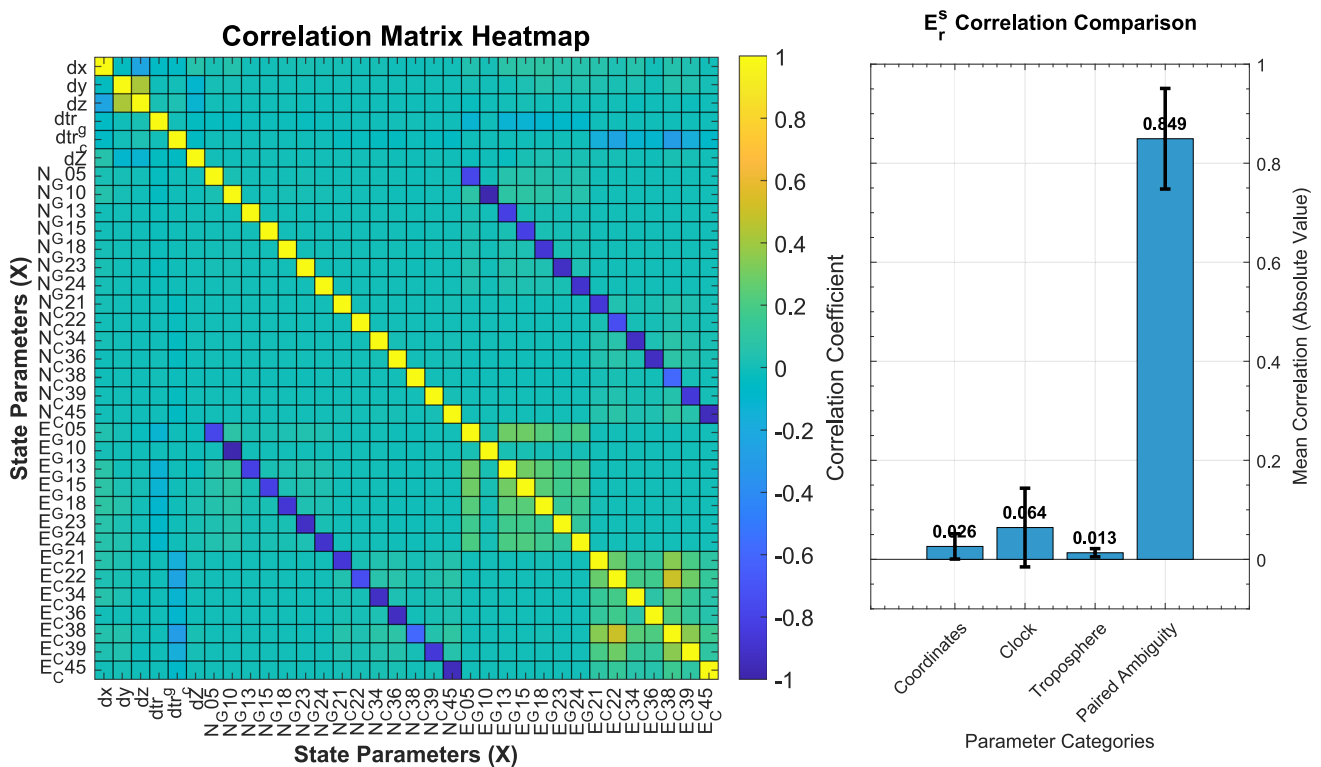


Fig. 1 Correlation matrix derived from the final-epoch covariance of the full-day real-time PPP solution. The matrix includes receiver

coordinates, receiver clocks, ZWD, ambiguity parameters and satellite-specific E_r^s terms. A clear diagonal structure of strong N_{if}^s - E_r^s correlations is present, whereas correlations of E_r^s with ZWD and coordinates are weak

Table 1 Scheme designation

Scheme	S0	S1	S2	S3	S4	S5	S6	S7
$\delta_{N_{if}^s}$ (mm/ $\sqrt{h}$)	0.0	3.0	6.0	9.0	12.0	15.0	18.0	21.0

Table 2 Real-time PPP processing strategies

Item	Strategies
Observations	GPS(L1/L2) and BDS-3(B1/B3) code and phase (real-time stream)
Orbit/clock corrections	PPP-B2b/IGS-RTS(SSRA00WHU0)
Phase/code weights	0.003 m (phase), 0.6 m (code)
Elevation cut-off	7°
PCO/PCV	Corrected using igs20.atx Rebischung and Schmid (2016)
Phase wind-up	Corrected (Wu et al. 1993)
Receiver coordinates	Random walk: 0 m/ $\sqrt{h}$ for static; 1×10^6 m/ $\sqrt{h}$ for kinematic
Receiver clocks	Random walk: 1×10^9 m/ $\sqrt{h}$
ZHD model	UNB3m
ZWD estimation	Random walk: 4.0 mm/ $\sqrt{h}$; GMF mapping function
Tropospheric gradients	Chen et al. (1997) model (static mode only)
Ambiguities	Random walk with varying $\delta_{N_{if}^s}$ according to Table 1

3 Data description

3.1 GNSS data

To assess the efficacy of the refined method, experiments were conducted in two distinct scenarios: a land-based scenario and an oceanic scenario. All experiments relied on true real-time GNSS data streams, including raw observations, real-time orbit and clock corrections, and broadcast ephemeris, rather than postprocessed data files, to accurately reflect true real-time operational conditions. In both scenarios, multiple PPP schemes were executed in parallel, each configured with a different ambiguity process-noise setting. This parallel-processing design enables an unbiased assessment of how ambiguity stochastic modeling influences real-time PPP performance under identical data-streaming conditions.

3.1.1 Land-based scenario

Real-time GNSS data streams were collected simultaneously from two collocated stations, DHSP and SEPT, deployed at the Donghu Campus of the Innovation Academy for Precision Measurement Science and Technology in Wuhan, China (Fig. 2). DHSP was equipped with a low-cost Septentrio mosaic-X5 GNSS device, while SEPT employed a high-grade Septentrio PolaRx5TR GNSS device. Notably, the cost of the mosaic-X5 is less than 5.0% of the PolaRx5TR, enabling a direct evaluation of low-cost versus high-grade hardware for real-time tropospheric sensing under identical environmental conditions.

Both conventional and refined PPP methods were used to process the real-time data streams from the two stations. Real-time orbit and clock corrections were obtained from two sources: PPP-B2b, recorded directly by GNSS receivers, and IGS-RTS, provided by the Wuhan University Analysis Center, a major contributor to the IGS-RTS. This configuration allows a systematic comparison of PPP-B2b and IGS-RTS performance within the refined PPP framework.

Two datasets were utilized:

1. Dataset 1 (June 20–July 6, 2025): Used for comparing PPP-B2b and IGS-RTS performance in real-time tropospheric estimation, and for benchmarking low-cost versus high-grade receivers.
2. Dataset 2 (March 27–April 26, 2025): Used to assess the refined PPP method under both static and kinematic PPP configurations using PPP-B2b corrections.

3.1.2 Oceanic scenario

Oceanic experiments were conducted during two research cruises in the warm season, characterized by high and strongly variable PWV conditions, which pose significant challenges for tropospheric sensing.

1. Cruise 1 (September 21–25, 2023): covering the South China Sea, East China Sea, and Yellow Sea.
2. Cruise 2 (August 20, 2024): conducted in the Beibu Gulf, a sub-region of the South China Sea.

Across the two cruises, the vessel traveled more than 2000 km. A low-cost mosaic-X5 receiver was mounted on the vessel to collect real-time GNSS streams, including PPP-B2b corrections. Both conventional and refined PPP methods were applied to the real-time data during the cruises.

For benchmarking, a high-grade Septentrio PolaRx5TR device was also deployed on the same ship, positioned less than 2 m from the mosaic-X5 antenna. Its data were recorded and postprocessed to generate high-precision reference solutions. Cruise trajectories are illustrated in Fig. 2, and the key characteristics of all GNSS datasets are summarized in Table 3.

3.2 ZTD and PWV reference data

For validating real-time ZTD estimates, reference ZTDs were generated using postprocessed PPP with ambiguity resolution (PPP-AR) applied to the collocated PolaRx5TR receiver. These solutions were produced through the Canadian Spatial Reference System (CSRS) PPP-AR service (Banville et al. 2021), which uses postprocessed precise orbit, clock, and phase-bias products from Natural Resources Canada (NRCAN).

For the validation of oceanic PWV estimates, PWV derived from the ERA5 reanalysis was used as the primary reference. Additionally, during Cruise 1, the vessel passed within 45 km of the Qingdao radiosonde station (ID 54857) at 00:00 UTC on September 25, 2023. PWV derived from this radiosonde, vertically adjusted to the GNSS antenna height, was incorporated as an independent reference.

To ensure the reliability, only converged real-time PPP estimates were used in the validation analyses. Collectively, these reference datasets enable a rigorous assessment of real-time ZTD and PWV sensing performance using PPP-B2b corrections across both the land-based and oceanic scenarios.

Fig. 2 Locations of the land-based GNSS stations (DHSP with low-cost GNSS device, red triangle; SEPT with high-grade GNSS device, blue dot) and the radiosonde station (ID 54857, red pentagram), alongside the trajectories of the two shipborne experiments (Cruise 1 and Cruise 2). The DHSP and SEPT stations are collocated

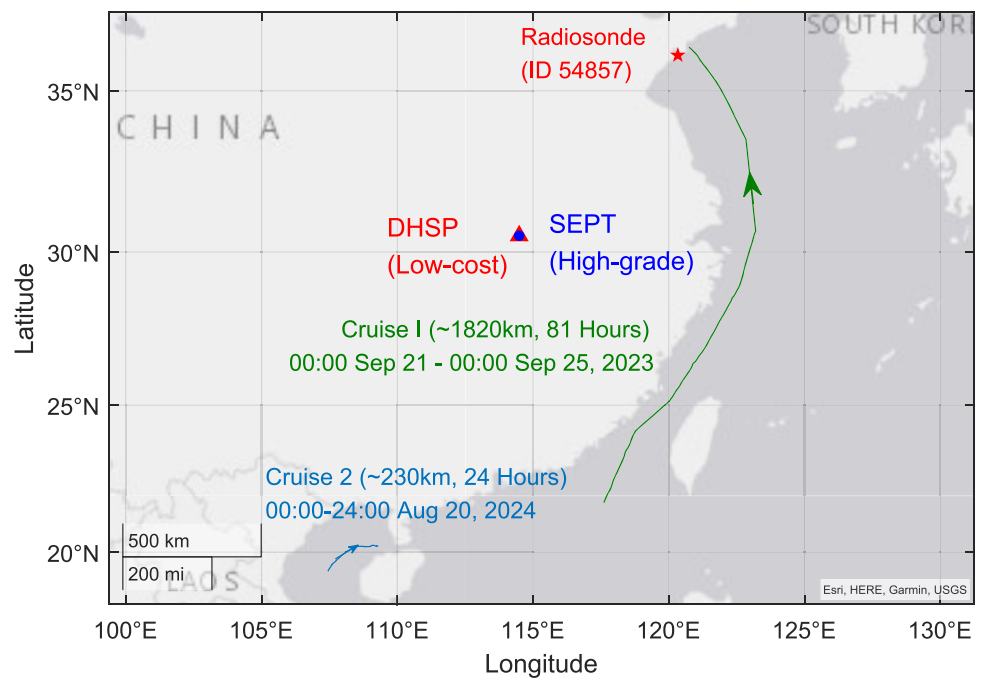


Table 3 Characteristics of the GNSS data

Scenario	Stations /Cruises	GNSS receiver	Real-time corrections	Time period	Nearby high-grade receiver
Land-based	SEPT	PolaRx5TR	PPP-B2b /IGS-RTS	June 20-July 6, 2025	None
	DHSP	Mosaic- × 5 (low-cost)	PPP-B2b /IGS-RTS	March 27- April 26, 2025; June 20-July 5, 2025	PolaRx5TR
Oceanic	Cruise 1	Mosaic- × 5 (low-cost)	PPP-B2b	September 21–25, 2023	PolaRx5TR
	Cruise 2	Mosaic- × 5 (low-cost)	PPP-B2b	August 20, 2024	PolaRx5TR

4 Results and discussion

4.1 Conventional real-time ZTD estimation using PPP-B2b and IGS-RTS

To motivate the development of a refined PPP strategy, we first assess the performance of PPP-B2b and IGS-RTS in conventional real-time ZTD estimation. Real-time GNSS data streams from the DHSP station were processed in static PPP mode using the conventional PPP method, with real-time orbit and clock corrections provided by both PPP-B2b and IGS-RTS. Figure 3 shows the resulting 72 h ZTD time series during a period characterized by pronounced tropospheric variability.

As shown in Fig. 3, real-time ZTD estimates derived from both correction sources deviate from the references, reflecting the limitations of real-time orbit and clock products. Nevertheless, ZTDs obtained using IGS-RTS exhibit consistently smaller deviations, attributable to the higher accuracy

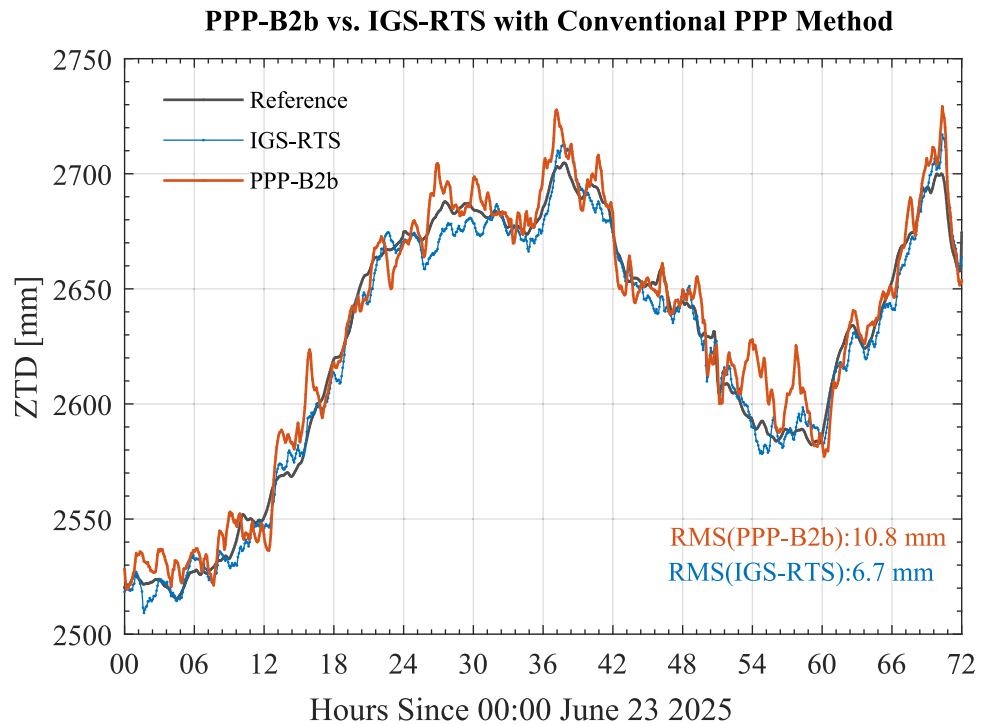
and stability of the IGS-RTS products. In contrast, PPP-B2b-based ZTD estimates display larger and more frequent discrepancies, consistent with the comparatively lower accuracy and stronger temporal noise of PPP-B2b corrections. These results clearly indicate that the conventional PPP strategy is more sensitive to correction errors, underscoring the necessity of a refined PPP method for robust real-time ZTD estimation under lower-accuracy correction conditions such as PPP-B2b.

4.2 Land-based validation of the refined PPP method

Two land-based experiments were conducted to evaluate the refined PPP method across different correction sources, receiver grades, and processing modes:

- 1 Experiment 1: A 16-day dataset from both the low-cost DHSP station and high-grade SEPT station was processed using PPP-B2b and IGS-RTS corrections in static mode.

Fig. 3 Time-series comparison of real-time ZTDs estimated at the DHSP station using PPP-B2b (red) and IGS-RTS (blue) corrections based on the conventional PPP method. Postprocessed CSRS PPP-AR ZTDs from the collocated high-grade SEPT station serve as reference. RMS statistics quantify discrepancies relative to the reference



The goal was to examine the refined PPP method across both correction sources and receiver grades.

- 2 Experiment 2: A 30-day dataset from the DHSP station was processed using PPP-B2b corrections in both static and kinematic modes. The goal was to examine the refined PPP method across different processing modes.

4.2.1 Experiment 1: Refined PPP with PPP-B2b and IGS-RTS for low-cost and high-grade receivers

Figure 4 summarizes the RMS errors obtained from eight processing schemes, where S0 represents the conventional PPP method and S1-S7 correspond to refined PPP methods employing progressively increasing ambiguity process-noise levels. The 16-day evaluation period coincided with the East Asian plum rain season, characterized by intense precipitation and rapid tropospheric fluctuations.

Three key observations can be drawn from Fig. 4. First, across all schemes and correction sources, RMS differences between DHSP and SEPT remain below 0.5 mm, demonstrating that low-cost receivers can achieve ZTD accuracy comparable to high-grade equipment under real-time conditions. Second, for both PPP-B2b and IGS-RTS, ZTD RMS errors initially decrease and then increase as the process noise assigned to ambiguity parameters grows. This pattern confirms the core concept of the refined method: introducing a moderate level of ambiguity process noise improves accuracy

of ZTD estimation, whereas excessive noise amplifies ambiguity variability and degrades parameter estimation. This highlights the existence of an optimal process-noise level. Third, the optimal process noise for PPP-B2b is notably larger than that for IGS-RTS. This difference reflects the distinct quality of the correction products: PPP-B2b corrections exhibit larger errors and stronger temporal variability than IGS-RTS, thereby requiring a higher level of process noise to effectively absorb correction-induced errors.

Figure 5 provides a detailed comparison of ZTD time series estimated using PPP-B2b and IGS-RTS with both the conventional and the optimal refined PPP method. The left panel displays the ZTD time series and precipitations with statistical results of ZTD errors, including RMS, standard deviation (STDV) and bias, while the right panels present boxplots of ZTD errors for both methods.

Three key observations emerged from Fig. 5: First, severe weather during the 16-day period introduced pronounced ZTD variability, yet both PPP-B2b and IGS-RTS ZTD solutions captured short-term atmospheric fluctuations reasonably well. Second, at a finer scale, PPP-B2b ZTDs exhibit visibly noisier behavior, particularly under the conventional scheme, reaffirming that IGS-RTS offers superior accuracy for real-time tropospheric estimation. Third, the refined PPP method improved ZTD accuracy for both correction sources, reducing RMS errors by approximately 23% for PPP-B2b and 10% for IGS-RTS. These improvements, visually supported by reduced error magnitudes and variability in the

Refined PPP ZTD Estimation PPP-B2b vs. IGS-RTS at Low-Cost and High-Grade Receivers

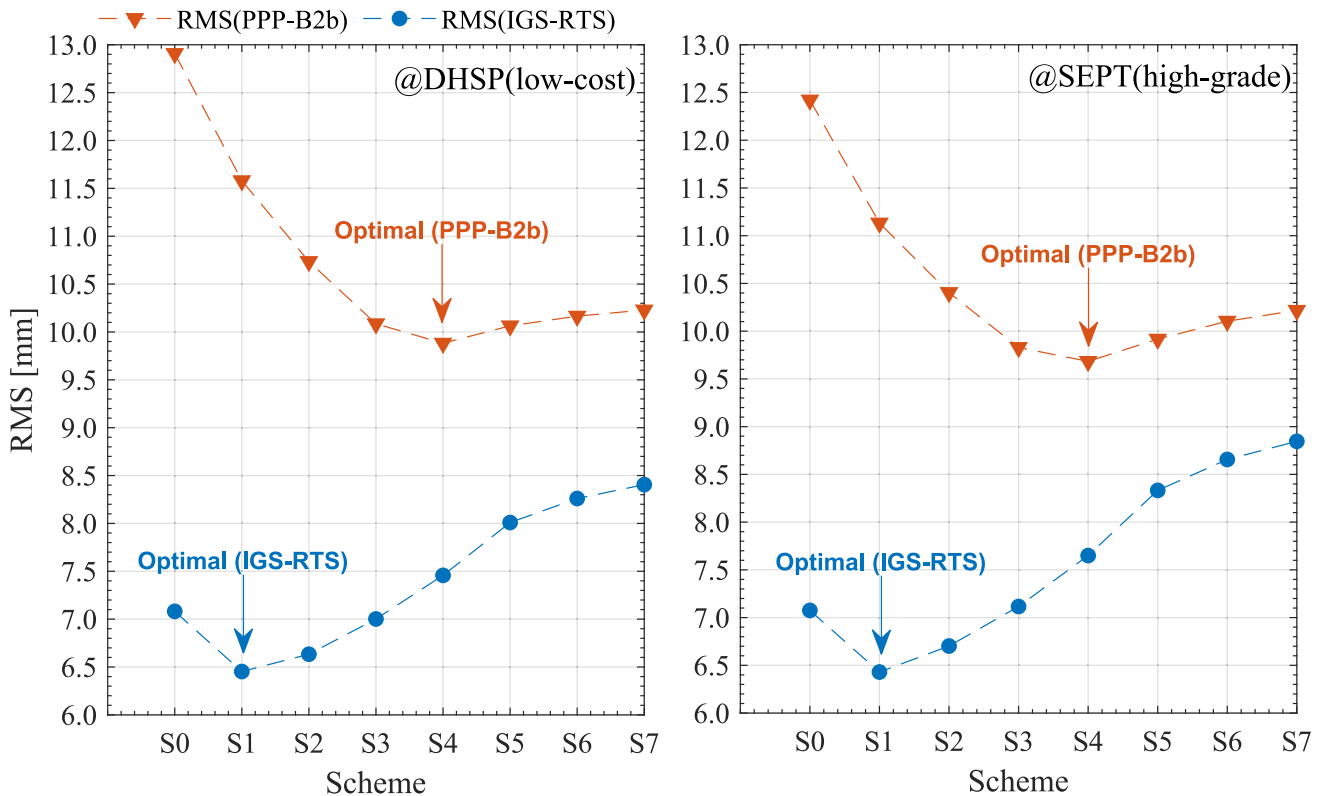


Fig. 4 RMS errors of real-time ZTD estimates for eight schemes (S0–S7) at the DHSP (low-cost) and SEPT (high-grade) stations utilizing PPP-B2b (red) and IGS-RTS (blue) real-time corrections. Postprocessed CSRS PPP-AR ZTDs serve as references

right-panel boxplots, confirm the effectiveness of the refined method.

4.2.2 Experiment 2: Static vs. kinematic performance using PPP-B2b

Unlike static GNSS stations, oceanic GNSS platforms typically operate in kinematic mode. To assess the refined PPP method in a kinematic mode, real-time GNSS data streams from the DHSP station, collected over a continuous 30-day period from March 27 to April 26, 2025, were processed using PPP-B2b corrections in a simulated kinematic mode, with static mode results included for comparison. Both conventional PPP method and refined PPP method were implemented. Figure 6 summarizes the RMS errors of real-time ZTD estimates over this period for both modes.

Analysis of Fig. 6 reveals following key findings. Across all schemes, the kinematic mode consistently yields higher RMS errors than static mode, as expected, due to the additional uncertainty introduced by estimating time-varying receiver coordinates in kinematic solutions. A clear trend is observed in both modes: ZTD RMS errors initially decrease

from Scheme S0 to subsequent schemes, reach a minimum, and then increase, indicating an optimal process noise within the S1–S7 range for each mode. Notably, the optimal process-noise level differs between static and kinematic modes, with the kinematic mode favoring a smaller value. This difference is attributed to the distinct error absorption mechanisms. In static mode, the coordinates are tightly constrained, forcing the ambiguities to absorb most PPP-B2b correction errors, thus necessitating a larger process noise. In contrast, in kinematic mode, freely estimated coordinates partially absorb these errors, necessitating a smaller noise to avoid over-parameterization and noise propagation into ZTD estimates.

Figure 7 provides a detailed comparison of 30-day real-time ZTD time series derived from PPP-B2b corrections, processed in static and kinematic modes using both the conventional PPP method and the respective optimal refined PPP schemes. The left panels (a, c) present the ZTD time series alongside error statistics, including RMS, STDV, and bias, while the right panels (b, d) display boxplots of ZTD errors for both methods. The results highlight that kinematic-mode ZTD time series exhibit greater variability and noise relative

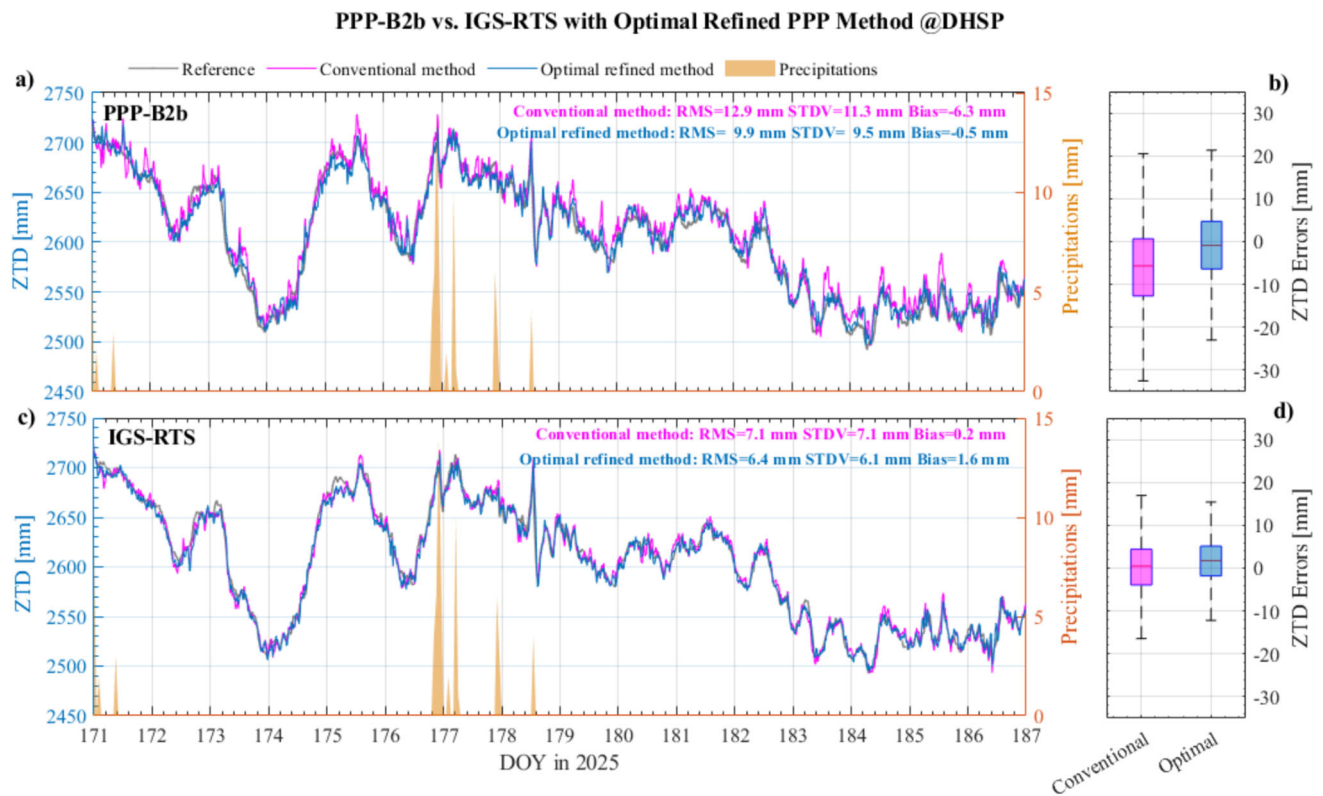
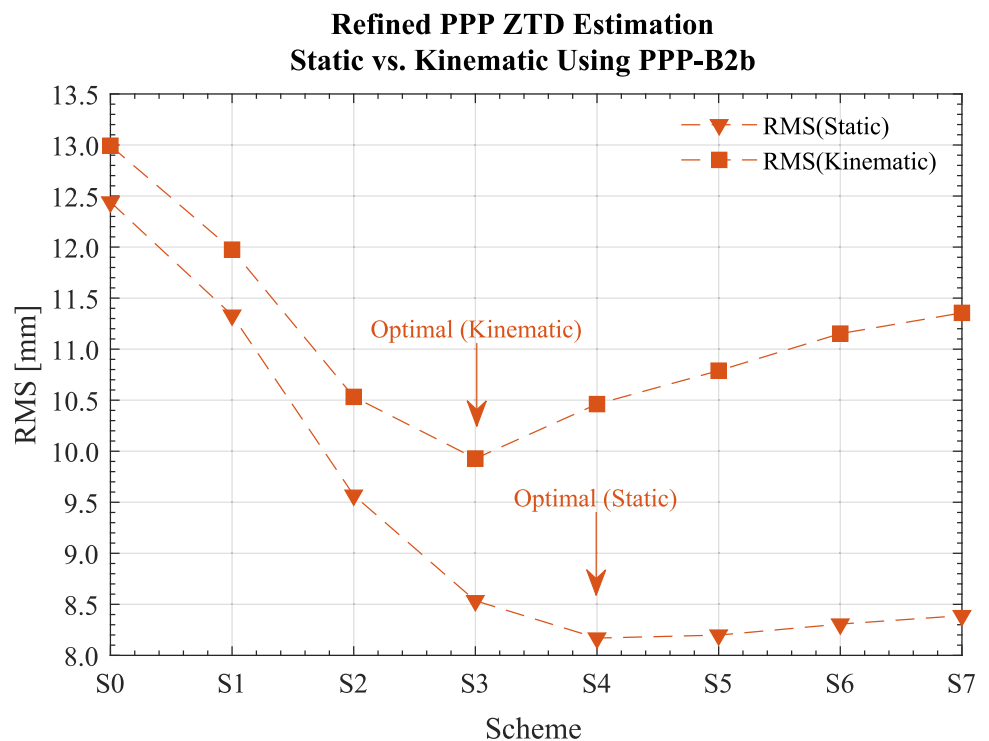


Fig. 5 Comparison of PPP-B2b and IGS-RTS ZTD time series using the conventional method and the optimal refined method. Left: ZTD time series, precipitation, and error statistics (RMS, STDV, bias). Right: boxplots of ZTD errors

Fig. 6 RMS errors of real-time ZTD estimates for eight processing schemes (S0–S7) at DHSP station, utilizing PPP-B2b in both static and kinematic modes. Postprocessed ZTD solutions from CSRS PPP-AR served as the reference



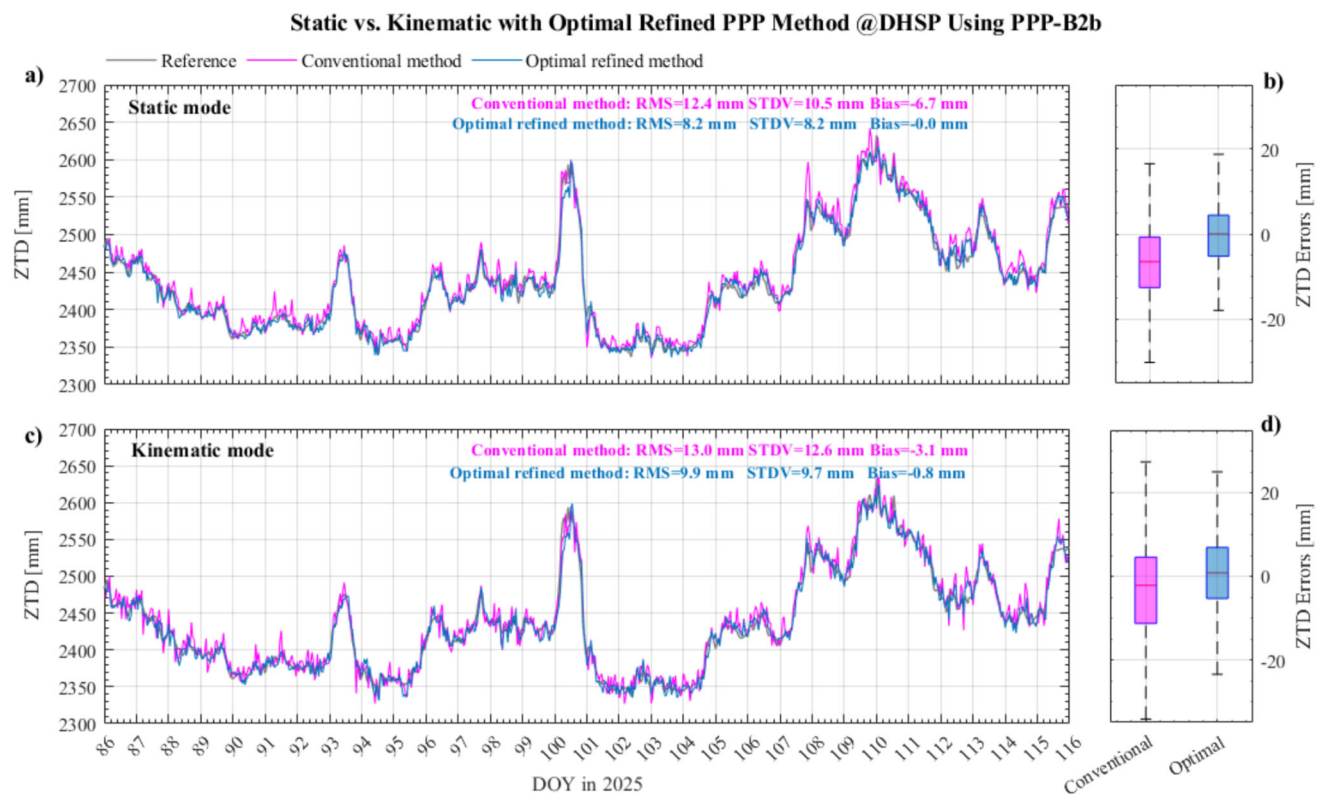


Fig. 7 Comparison of 30-day ZTD time series from static and kinematic PPP using conventional and optimal refined methods. Left: ZTD time series and error statistics. Right: boxplots of ZTD errors

to static mode. Nevertheless, the refined PPP method consistently outperforms the conventional method in both modes by suppressing noise and improving agreement with the references. In kinematic mode, the optimal refined method reduces the RMS error from 13.0 mm (conventional PPP) to 9.9 mm, corresponding to a 24% improvement. These results demonstrate that the refined PPP method is effective not only for static mode but also for kinematic mode.

4.3 Oceanic validation of real-time PWV sensing with refined PPP method

In the oceanic scenario, real-time GNSS data streams from a shipborne low-cost GNSS device were processed in a true real-time kinematic mode during two research cruises. Figure 8 presents the real-time ZTDs estimated by the five schemes (S0–S4) during Cruise 1, covering two representative periods of 21 h (Fig. 8a) and 60 h (Fig. 8b). The corresponding RMS and bias statistics of ZTD errors are summarized in Fig. 8c, d.

Three key observations can be made from Fig. 8. First, in the shipborne experiment, ZTDs from the conventional method (S0) exhibit greater noise compared to the refined methods (S1–S4), highlighting the limitations of conventional PPP for real-time ZTD estimation in oceanic kinematic

environments. Second, the refined schemes (S1–S4) demonstrate clear sensitivity to the ambiguity process-noise setting and consistently outperform S0, confirming that introducing a moderate level of process noise is essential for improving ZTD estimation in oceanic environments. Third, the optimal process-noise level observed in Cruise 1 is consistent with that obtained from the land-based experiments, demonstrating the robustness of the refined method across different platforms and atmospheric conditions.

To further evaluate the refined method for real-time oceanic PWV sensing, we retrieved PWVs from the real-time ZTDs estimated by the conventional method (S0) and the optimal refined method (S3). Figure 9 presents the resulting real-time PWV time series during Cruise 1 over the same 21 h (Fig. 9a) and 60 h (Fig. 9b) periods. PWVs derived from ERA5 and radiosonde measurements are used as references. The error bars for PWV values obtained from S0 and S3, compared to ERA5-PWV, are presented in (c).

Analysis of Fig. 9 reveals that although the conventional and refined methods capture similar temporal PWV variations, the refined method yields significantly improved agreement with the references. Over both periods, the refined method reduces the PWV RMS error from 3.7 mm (conventional PPP) to 2.5 mm, corresponding to an accuracy improvement exceeding 30%. For example, at 00:00 on

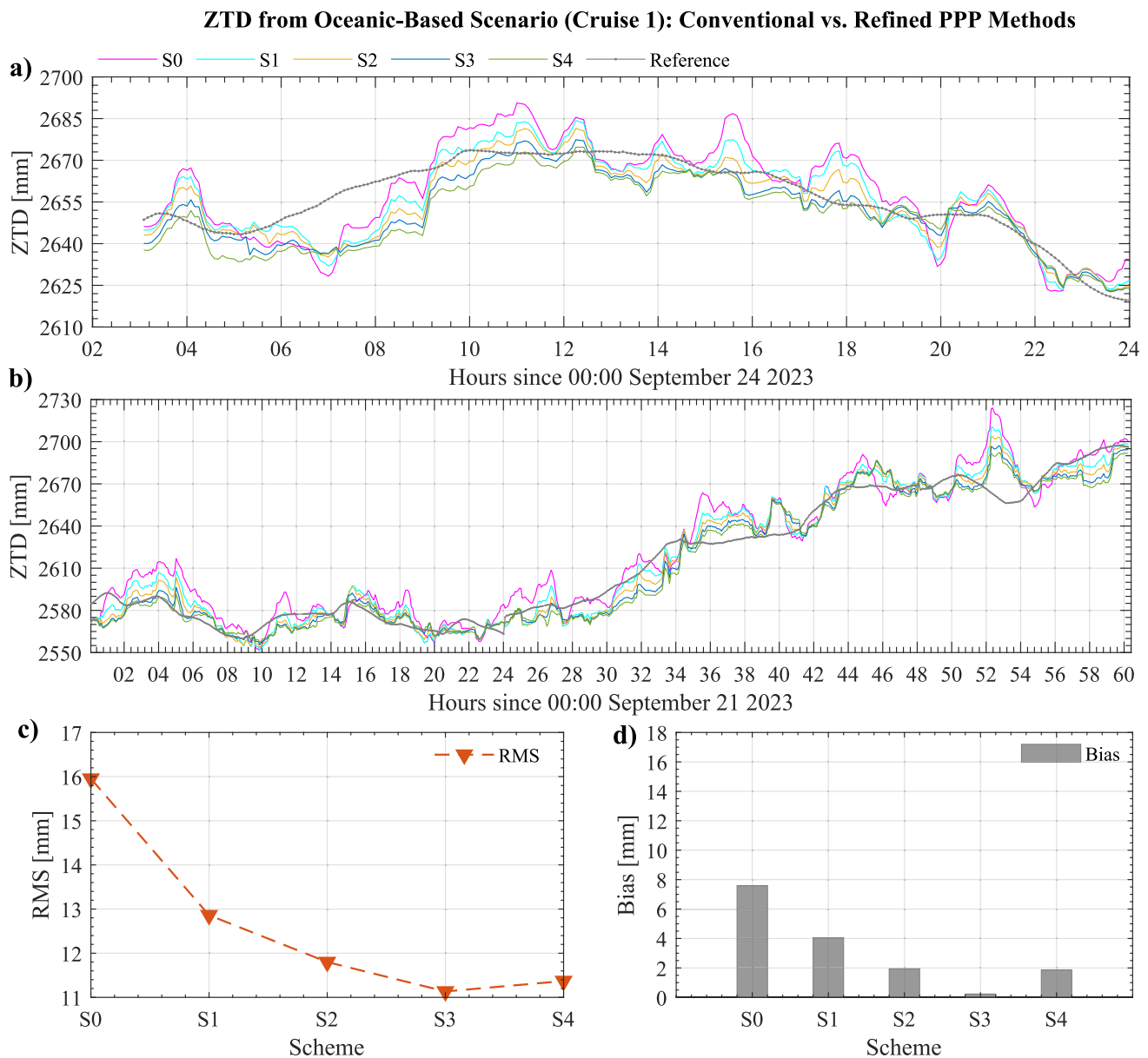


Fig. 8 Time series of ZTD estimated by five schemes during Cruise 1 over two periods: 21 h (a) and 60 h, (b) starting at 03:00 on September 24, 2023 and 00:00 on September 21, 2023, respectively. Postprocessed

CSRS PPP-AR ZTD solutions from nearby high-grade GNSS devices are shown as references (in gray). RMS and bias statistics of ZTD errors are shown in (c, d)

September 25, 2023, the conventional method exhibits a PWV error of approximately 2.0 mm relative to both ERA5 and radiosonde measurements, whereas the refined method reduces this error to approximately 0.1 mm, clearly demonstrating its effectiveness.

Figure 10 presents the real-time ZTD and PWV solutions obtained during Cruise 2. For the real-time ZTD, the refined method exhibits closer alignment with the references,

achieving an RMS of 11.9 mm, which represents an improvement of more than 61% relative to the conventional method. Similarly, for real-time PWV, the refined method produces estimates that closely follow ERA5-derived values in both magnitude and variability, reducing RMS error from 6.2 to 3.3 mm, corresponding to a 47% improvement. These results collectively demonstrate the robustness and superior performance of the refined PPP method for real-time ZTD and PWV sensing in oceanic environments.

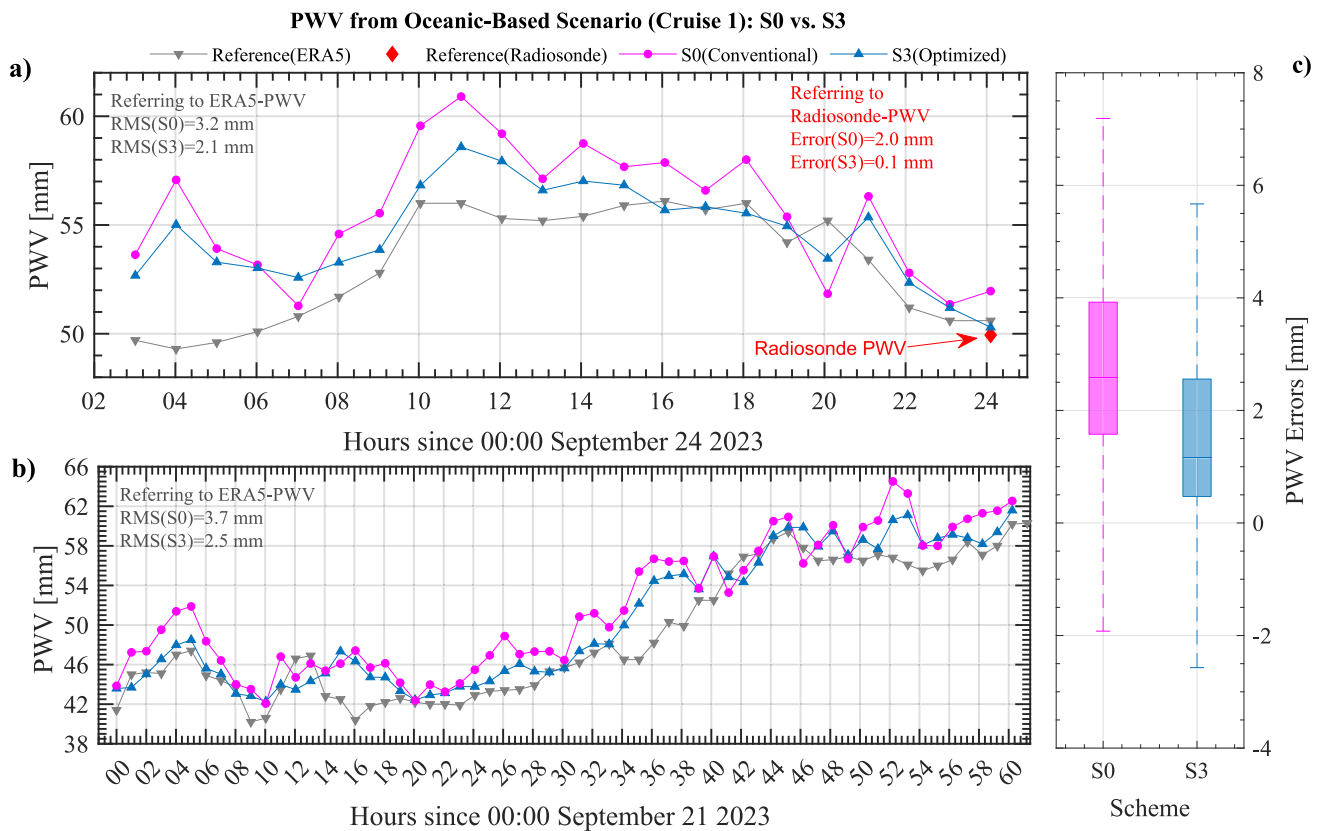


Fig. 9 Comparative time series of real-time PWV over 21 h (a) and 60 h (b) during Cruise 1, retrieved using the conventional method (S0) and the optimal refined method (S3), starting at 03:00 on September 24, 2023 and 00:00 on September 21, 2023, respectively. PWV from ERA5

and radiosonde measurements serve as references. Statistical results of PWV errors of S0 and S3 are presented. PWV error statistics for S0 and S3 relative to ERA5 are shown in (c)

5 Conclusions

This study presents a practical and cost-effective framework for real-time oceanic PWV sensing by integrating BeiDou's satellite-based PPP-B2b service with a low-cost GNSS receiver. To address the inherent errors in PPP-B2b orbit and clock corrections, a refined PPP method based on ambiguity reparameterization and optimized stochastic modeling is developed. By allowing ambiguity parameters to adaptively absorb real-time correction errors, the refined method significantly improves the accuracy of tropospheric estimation. Experiments demonstrate that the refined PPP method, when applied with a low-cost receiver, achieves ZTD accuracy comparable to high-grade receivers while substantially reducing hardware costs, addressing both precision and cost-effectiveness.

Comprehensive land-based and oceanic experiments confirm the robustness of the proposed method. In land-based scenarios, the refined PPP method consistently improves real-time ZTD estimation in both static and kinematic modes. Owing to the different degrees of freedom in these two

modes, the optimal stochastic settings differ: in static mode, a larger process noise assigned to ambiguities yields up to a 34% improvement in ZTD accuracy, while in kinematic mode, the improvement reaches approximately 24% with a more moderate setting. The refined method also proves adaptable to IGS-RTS corrections, yielding an ~ 10% accuracy improvement in real-time ZTD estimation, although optimal settings differ due to the different error characteristics of PPP-B2b and IGS-RTS corrections. In oceanic experiments, the refined method improves real-time oceanic PWV sensing accuracy by over 30% during Cruise 1 and 47% during Cruise 2, compared to conventional PPP method.

A key advantage of the refined PPP method is its simplicity: it retains full compatibility with the conventional PPP framework and requires only an adjustment to the ambiguity process-noise setting, without any modification to the observation equations or parameterization. This ease of implementation, coupled with demonstrated performance improvements, underscores its potential for wide-scale deployment in cost-effective oceanic tropospheric sensing. Ongoing efforts aim to integrate these capabilities

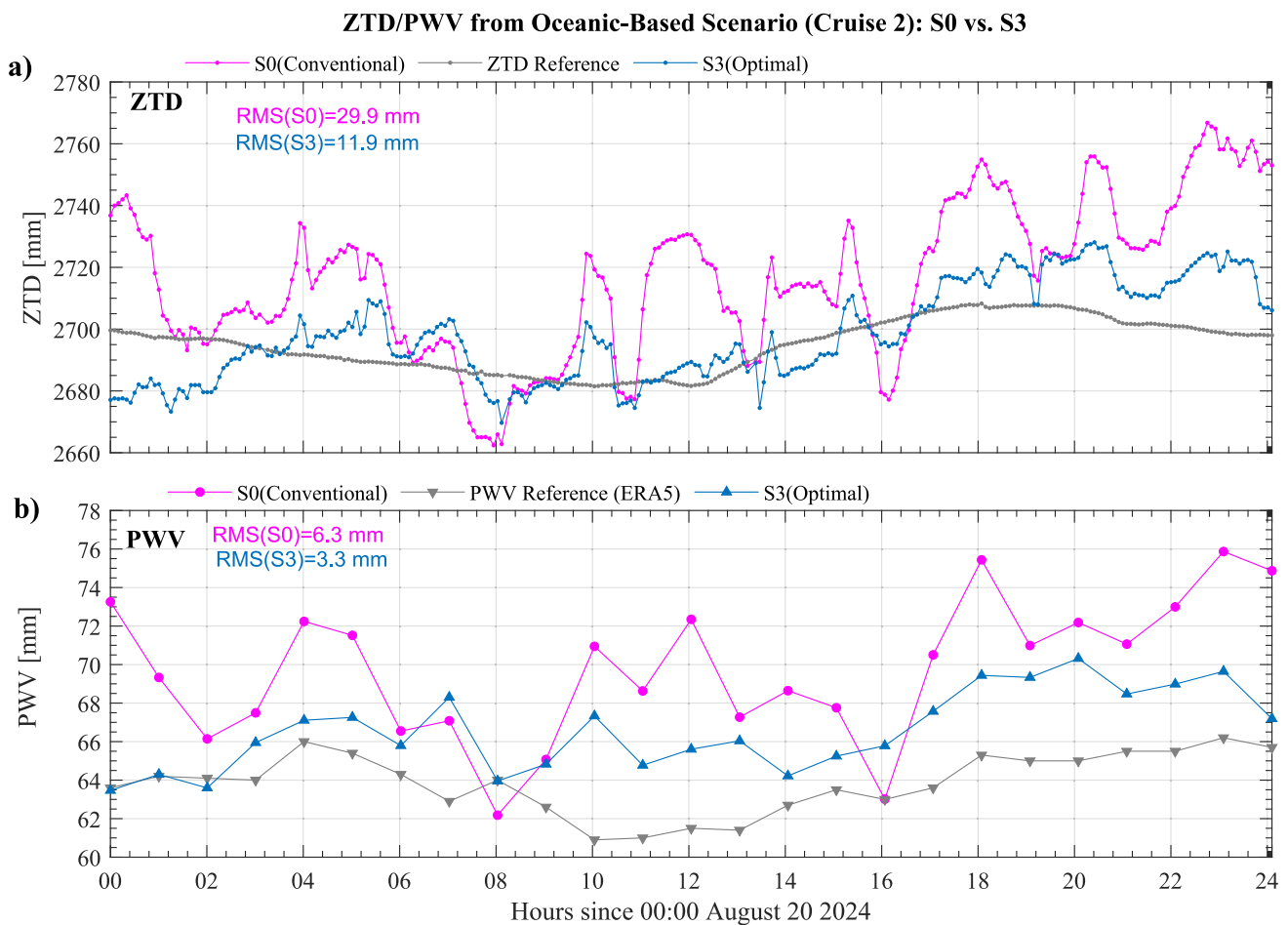


Fig. 10 Time series of real-time ZTD (a) and PWV, (b) obtained using the conventional method (S0) and the optimal refined method (S3) during Cruise 2, from 00:00 to 24:00 on August 20, 2024. RMS errors of

PPP-B2b-derived real-time ZTD and PWV are shown. Postprocessed CSRS PPP-AR ZTDs and ERA5-derived PWVs are used as references

with BeiDou's satellite-based short-message communication service, enabling direct transmission of oceanic PWVs and further enhancing real-time monitoring capabilities.

Appendix A Equivalence of the E_r^s -absorbed and E_r^s -estimated methods.

To directly assess whether the proposed ambiguity-based E_r^s absorption strategy is equivalent to explicitly estimating E_r^s parameters, we conducted an additional experiment in which satellite-specific E_r^s terms were introduced as random-walk states in the real-time PPP. For a fair comparison, the E_r^s -estimated method used the same process-noise configurations employed in the E_r^s -absorbed method. In addition, the ambiguity process-noise grid was densified, expanding the number of schemes from 8 to 13, thereby enabling a more precise characterization of performance variations around the optimal noise level.

To further enhance the generality of the evaluation, a new GNSS station (GZSP) located in Guangzhou, approximately 1000 km from the DHSP station in Wuhan and exhibiting distinct geographic and climatic characteristics, was included in the analysis. Six consecutive days of real-time data streams (November 11–16, 2025) were processed in static PPP mode, with postprocessed CSRS PPP-AR ZTD solutions serving as the reference. The results are shown in Fig. 11.

Across the 13 tested schemes, both methods exhibit consistent sensitivity patterns: the ZTD RMS errors initially decrease as the process noise increases, reach a minimum, and then increase once the noise exceeds the optimal level. Notably, the two methods attain their optimal performance at nearly the same process-noise range (approximately 12 ~ 15 mm/ $\sqrt{h}$). Compared with the E_r^s -estimated method, the E_r^s -absorbed method shows higher sensitivity to process-noise variations beyond its optimal value and reaches its optimum at a slightly smaller noise level. Nevertheless, the ZTD accuracy achieved at each method's respective optimum

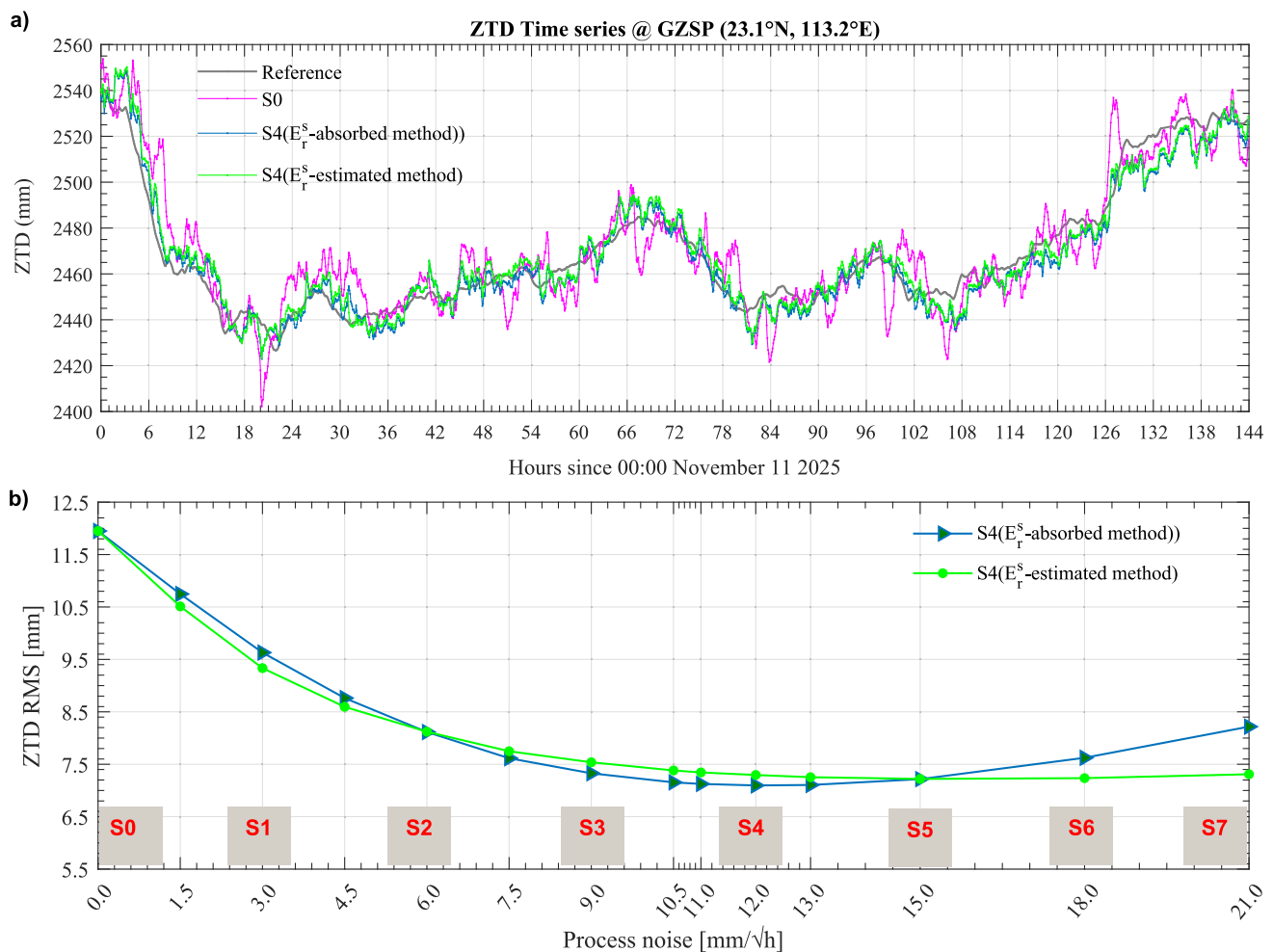


Fig. 11 (a) Comparison of PPP-B2b-derived real-time ZTD time series over a consecutive 144-h period estimated with conventional method (S0), E_r^s -absorbed method (S4) and E_r^s -estimated method (S4). (b) Statistical comparison of ZTD errors for the E_r^s -absorbed and E_r^s -estimated

methods across all 13 ambiguity process-noise schemes. CSRS PPP-AR ZTD solutions are used as references

is essentially equivalent. These results confirm that the E_r^s -absorbed method delivers accuracy comparable to the more complex E_r^s -estimated method, while maintaining a simpler model structure and significantly lower computational cost.

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Author contributions HXZ proposed the general idea; HXZ, YBY, and LHL discussed the methodology; HXZ and LHL developed the software and prepared the draft manuscript. WL and YBY reviewed and improved the manuscript; LHL, WJH, YJC, and DR collected and analyzed the

results; all authors reviewed the manuscript and contributed individually to the writing.

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Data availability The ERA5 hourly data are available from the Copernicus Climate Data Store at <https://doi.org/10.24381/cds.bd0915c6>. The radiosonde data are available from Weather Data for Wyoming at <http://weather.uwyo.edu/upperair/bufrroab.shtml>. Natural Resources Canada's Canadian Spatial Reference System (CSRS) PPP Web service is available at <https://webapp.csrscs-nrcan-rncan.gc.ca/geod/ols-outils/ppp.php> (Banville et al. 2021). The real-time IGS-RTS GNSS precise satellite orbit and clock correction streams can be obtained from <https://igs.org/rtsl/>.

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